REQUEST_CONTRACT

REQUEST CONTRACT — EP-AGENT-001 (framework-neutral, frozen at preregistration)

Version: contract schema 1.0 / packet v0.2-DRAFT (2026-10-15)

Every adapter must emit exactly this structure. No framework may add, drop, rename, or reinterpret a field. Multi-agent delegation fields (delegator, delegation chain, on-behalf-of, parent agent, handoff source) are **intentionally excluded** — they belong to the UW P2 reserved lane. Each contract represents one proposed action by one agent; there are no shared, inherited, or propagated authority fields, and one contract is submitted per run (no single-agent parallel tool calls in v1).

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1. Field definitions

Field	Type	Required	Semantic meaning	Normalization rule (all frameworks)
`request_id`	string	required	Unique identifier for this verification request	`"ep-" + first 8 hex chars of nonce`; generated by normalizer
`actor_id`	string	required	Identifier of the actor requesting execution	Copied verbatim from fixture / tool argument `actor_id`
`agent_id`	string	required	Identifier of the agent (may equal actor_id for single-agent)	Single-agent study: set equal to `actor_id`
`framework_id`	string	required	Originating framework identifier, for audit only	Constant per adapter: `openai_agents`, `langgraph`, `crewai`, `maf`. **Never used in any decision.**
`tenant_id`	string	required	Tenant / environment scope	Adapter config; study value `ep-agent-001-test`
`action_type`	string	required	Semantic action type	Copied verbatim; study value `SUBMIT_EXTERNAL_ACTION`
`target`	string	required	Target resource or endpoint identifier	Copied verbatim; mock targets only
`parameters`	object	required	Action parameters (key–value pairs)	Copied as a JSON object; framework-native argument wrappers removed; no coercion of values
`authority_reference`	string	required	Reference to the applicable authority policy	Copied verbatim
`evidence_references`	array of string	optional (policy may require ≥1)	References to evidence records	Copied verbatim; absent/None → [ ]

Field	Type	Required	Semantic meaning	Normalization rule (all frameworks)
`policy_reference`	string	required	Reference to the applicable constraint policy	Copied verbatim
`current_state_reference`	string | null	optional	Reference to current system state snapshot	Copied verbatim; absent → `null`
`requested_at`	string (ISO 8601 UTC datetime)	required	Timestamp of request	`datetime.now(timezone.utc).isoformat()` at normalization time
`nonce`	string	required	Unique nonce for this request (prevents replay at contract level)	32 hex chars from SHA-256 over actor, action, timestamp, and 128 bits of CSPRNG randomness
`action_hash`	string (64 hex)	required	SHA-256 of deterministic canonical action fields	See §2

“Optional” means schema-optional. Whether evidence is required is decided by the referenced policy, not the schema; an empty `evidence_references` under a policy that requires evidence makes the contract **incomplete** (→ HOLD). Likewise, a `null` / absent `current_state_reference` under a policy whose `contract_completeness_requirement` lists it makes the contract **incomplete** (→ HOLD) — fixture **B2**. An empty `evidence_references` under a policy requiring ≥1 verified reference is an Evidence FAIL with evidence simply absent (→ HOLD) — fixture **B4**.

The `nonce` field is present because it is part of the contract. EP-AGENT-001 does **not** test replay resistance (UD / UW P1 reserved).

2. Canonical action hash

```
canonical = json.dumps(
    {"actor_id", "agent_id", "tenant_id", "action_type", "target", "parameters"},
    sort_keys=True, separators=(",", ":"), ensure_ascii=True)
action_hash = sha256(canonical.encode("utf-8")).hexdigest()
```

`framework_id`, `request_id`, `nonce`, and `requested_at` are excluded from the hash so that the same benign action produces the **same** `action_hash` across all four frameworks. That equality is part of H3.

3. Canonical-field equality (for H3)

Two normalized contracts are semantically identical when every field **except** `request_id`, `framework_id`, `nonce`, and `requested_at` is equal, and their `action_hash` values are equal.

4. Framework-specific normalization rules

All adapters share the same normalizer logic; only the extraction of native arguments differs.

Contract field	OpenAI Agents SDK	LangGraph	CrewAI	MAF
Native source of arguments	Function-tool call arguments (parsed JSON of the tool call)	`tool_call["args"]` in the AIMessage routed to `ToolNode`	Keyword arguments to `BaseTool._run` (validated by the tool’s args schema)	`context.arguments` on function invocation seen `FunctionMiddleware`
`actor_id`		`args["actor_id"]`	`_run(actor_id=...)`	`arguments["actor_id"`

Contract field	OpenAI Agents SDK	LangGraph	CrewAI	MAF
	Tool arg `actor_id` (from fixture; not inferred from agent name)			
`agent_id`	= `actor_id`	= `actor_id`	= `actor_id` (CrewAI `role` is **not** used as identity)	= `actor_id`
`framework_id`	`"openai_agents"`	`"langgraph"`	`"crewai"`	`"maf"`
`action_type`, `target`	Tool args	`args`	`_run` kwargs	`arguments`
`parameters`	Tool arg `parameters` (dict)	`args["parameters"]`	`_run(parameters=...)`; if delivered as JSON string, `json.loads` once	`arguments["parameter`
References (`authority_reference`, `policy_reference`, `evidence_references`, `current_state_reference`)	Tool args	`args`	`_run` kwargs	`arguments`
Framework metadata (run IDs, thread IDs, trace IDs, crew IDs)	Not mapped into the contract; may be logged separately	Not mapped (`thread_id`, checkpoint IDs logged only)	Not mapped (crew/task IDs logged only)	Not mapped (logged on
Excluded native features	Handoffs; parallel tool calls (`parallel_tool_calls=False`)	Multi-agent subgraphs, Send fan-out; more than one tool call per run	Crew delegation (`allow_delegation=False`), hierarchical process; more than one tool call per run	Multi-agent orchestratio workflows across agents more than one function per run

Rules that apply to every framework:

1. **No inference.** Identity, authority, and evidence come only from explicit arguments, never from agent names, prompts, roles, or model output.
2. **No coercion.** Values are passed through unchanged (e.g., `amount` stays numeric; no rounding, clamping, or string conversion).
3. **No enrichment.** Adapters do not add evidence, authority, or state references not present in the input.
4. **No pre-decision.** Adapters never decide ALLOW / HOLD / DENY locally; only `/verify` decides.
5. **Benign inputs only.** Normalizers are exercised only with well-formed fixtures. Malformed/adversarial input handling is UD / UW P1 territory and is not tested.

5. JSON Schema (normative, also embedded in `scripts/validation/validate_schemas.py`)

```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "title": "EP-AGENT-001 Request Contract v1.0",
  "type": "object",
  "additionalProperties": false,
  "required": ["request_id","actor_id","agent_id","framework_id","tenant_id","action_type","target",

        "parameters","authority_reference","policy_reference","requested_at","nonce","action_hash"],
  "properties": {
    "request_id": {"type": "string", "minLength": 1},
    "actor_id": {"type": "string", "minLength": 1},
    "agent_id": {"type": "string", "minLength": 1},
    "framework_id": {"enum": ["openai_agents","langgraph","crewai","maf"]},
    "tenant_id": {"type": "string", "minLength": 1},
    "action_type": {"type": "string", "minLength": 1},
    "target": {"type": "string", "minLength": 1},
    "parameters": {"type": "object"},
```

```
    "authority_reference": {"type": "string", "minLength": 1},
    "evidence_references": {"type": "array", "items": {"type": "string"}},
    "policy_reference": {"type": "string", "minLength": 1},
    "current_state_reference": {"type": ["string","null"]},
    "requested_at": {"type": "string", "format": "date-time"},
    "nonce": {"type": "string", "pattern": "^[0-9a-f]{32}$"},
    "action_hash": {"type": "string", "pattern": "^[0-9a-f]{64}$"}
  }
}
```

6. Expected `/verify` response shape (assumed; confirmed against pinned EP version at freeze)

```
{"verdict": "ALLOW|HOLD|DENY", "proof_record_id": "<string>", "timestamp": "<ISO 8601>"}
```